

Organic Solvent and Surfactant Free Fluorescent Organic Nanoparticles by Laser Ablation of Aggregation-Induced Enhanced Emission Dyes

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Less toxic and highly fluorescent nanoparticles are in high demand to image biological events and early-stage disease. A strategy to fabricate highly fluorescent organic nanoparticles by laser ablation of aggregation-induced enhanced emission (AIE) luminophores, which are free of any organic solvent and surfactant, is presented. As these dyes provide no concentration quenching, the increased number of photoluminescent molecules in a nanoparticle produces bright fluorescence even with their small size (<2 nm), making them highly suitable for intracellular uptake and imaging for a variety of biomedical applications. The design and synthesis of a new AIE luminophore, DCeTDCS, are reported and its photoluminescent quantum yield enhancement up to 58% in the aggregated state is demonstrated. Extremely stable nanoparticles of this luminophore with a narrow size distribution, by laser ablation in water are reported and its superior optical properties that are comparable to quantum dots are verified. The highly negative surface charge of these nanoparticles impedes cellular uptake, but when the surface is coated with chitosan, a cationic polymer, intracellular uptake in microglia is achieved. The strategy provides a novel tool to produce in water, ultrasmall and surfactant-free highly fluorescent organic nanoparticles suitable for biomedical applications.

Photoluminescent nanoparticles such as quantum dots,^[1] nanophosphors,^[2] and fluorescent organic nanoparticles (FONs)^[3] have been key materials to image various biological events in vitro and in vivo. Inorganic nanoparticles (quantum dots and nanophosphors) have potential toxicity from the accumulation of heavy metals in living body.^[4] FONs are rather free from the issue of heavy metals, but their typical fabrication method, reprecipitation, can cause additional toxicity due to the use of toxic organic solvents,^[5] which are usually difficult to completely remove after mixing with water during the fabrication process. To minimize the toxicity of fluorescent nanoparticles, the development of a fabrication method without the use of organic solvents and surfactants is in great demand.

In this regard, pulsed laser ablation in water provides a promising alternative to conventional chemical synthesis routes.^[6] As already established in the literature, this

purely physical approach circumvents many limitations of chemical synthesis and can be performed under ultrapure conditions (e.g., deionized water, ethanol, etc.). In addition, when used in the ultrashort (femtosecond) laser ablation regime, this method allows the synthesis of an aqueous dispersion of ultrasmall nanoparticles, more suitable for many biological applications, even in the absence of chemically active substances.^[7] However, the concentration quenching effect encountered in the case of typical organic luminophores limits the application of this technique for making nanoformulations using laser ablation.

In this paper, we introduce a new class of organic luminophores called aggregation-induced emission (AIE) fluorophores as ideal materials for laser fabrication to produce highly fluorescent nanoparticles because they exhibit significantly enhanced fluorescence in aggregated states, which is in contrast to serious emission quenching for conventional luminophores.^[8] The concentration quenching nature of AIE luminophores (LPs) may be overcome by a variety of mechanisms including: i) restriction of nonradiative energy dissipation by rotational relaxation (e.g., tetraphenylethene);^[8] ii) inducing planarization to extend p-conjugation from twisted conformations (e.g., bis(styryl)anthracene derivatives);^[9] iii) head-to-tail arrangement of

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transition dipole moment (J-aggregation) (e.g., cyanostilbene derivatives);^[10] and iv) protecting intramolecular hydrogen bonds from interaction with solvent molecules (e.g., hydroxyphenyl benzoxazole)^[11] in the aggregated state. Unlike conventional FONs composing of a tiny amount of LPs and a large amount of matrix materials such as polymer and silica to avoid concentration quenching, AIE FONs can approach maximum absorption by increasing the quantities of LPs in the nanoparticle and, hence, achieve maximum brightness by composing nanoparticles of the pure LPs without any other substances.^[12]

Here, we demonstrate the feasibility of AIE FON fabrication by laser ablation to obtain bright nanoparticles devoid of organic solvent and surfactant, thus making them more suitable for bioimaging applications. Direct fabrication of FONs from the bulk powder of AIE LPs can avoid unexpected toxic interference originating from additional chemicals (i.e., surfactants) required to help disperse FONs in water. The AIE FONs showed a superior suspension stability within 2 nm size and a comparable brightness with quantum dot. With this AIE FON made by laser ablation, we successfully demonstrated a cellular imaging application after biocompatible modification using chitosan.

We designed and synthesized an AIE dicyanostilbene, DCeTDCS (Figure 1a), which combines two cyanostilbene units, a well-known AIE building block, by J-aggregation forming a head-to-tail arrangement of transition dipole moments.^[10] These types of dicyanostilbene (DCS) derivatives were previously reported for their tunable AIE emission property by switching their packing mode under specific stimuli such as pressure, heating, and solvent vapor.^[13] Because symmetric DCS derivatives have a symmetric transition dipole moment, they can make only partial head-to-tail packing (anti-parallel H-aggregation).^[13a] Hence, in contrast to J-aggregates generating a red-shifted absorption band, they have a blue-shifted absorption band originating from the repulsion of transition dipole moments, and the band can be shifted to a shorter wavelength with a more parallel arrangement of the dipole moments. Based on exciton splitting,^[14] the DCS derivatives had green photoluminescence (PL) at 533 nm by extended excited state delocalization, when they had partial antiparallel packing, while more parallel packing turns back the emission to blue at 458 nm by the loss of excited state delocalization.

Through simple two-step reactions, sequentially i) esterification of terephthalaldehydic acid and ii) Knoevenagel condensation with p-xylene dicyanide, we obtained DCeTDCS as a pale-yellow powder (Figure 1a). Under 365 nm UV lamp irradiation, DCeTDCS exhibits a bright green-yellow emission in the solid state. To evaluate DCeTDCS's AIE activity, we originally used a typical reprecipitation method. DCeTDCS was dissolved in

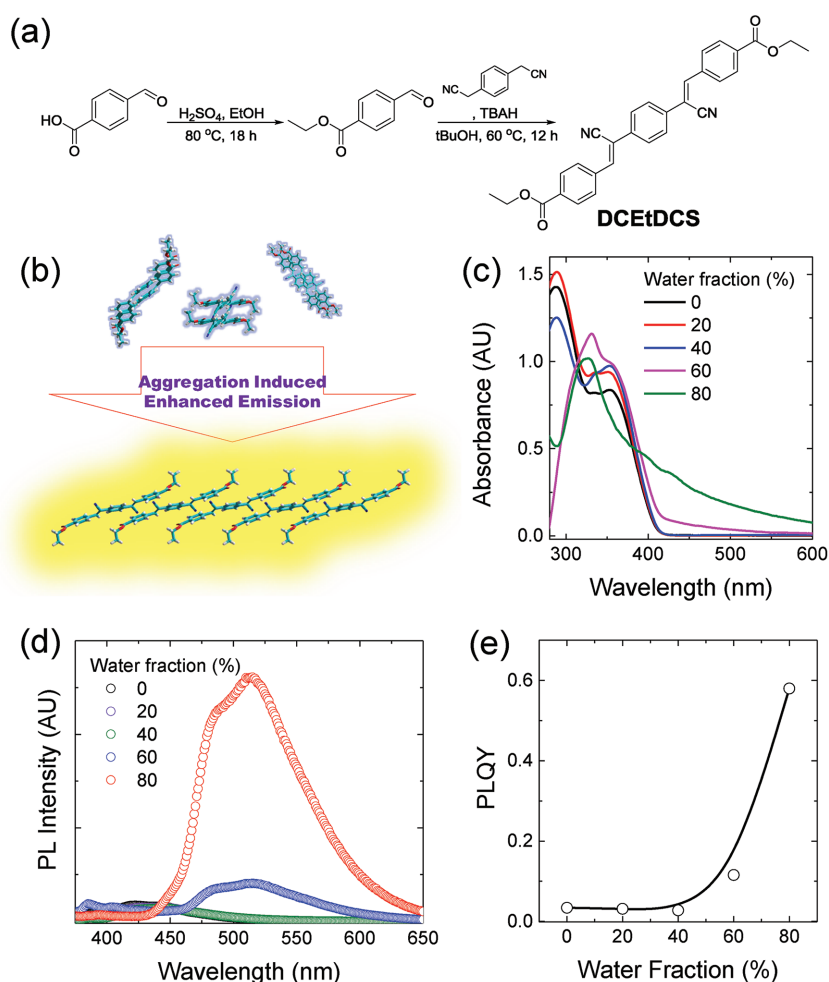


Figure 1. Synthesis and optical properties of DCeTDCS. a) Synthetic route of DCeTDCS. b) Schematic representation of AIE of DCeTDCS. c) Absorption and d) photoluminescence (PL) spectra of DCeTDCS (0.01 mg mL⁻¹) in THF/water mixed solvent. e) PLQY of DCeTDCS as a function of water fraction in THF/water mixed solvent.

tetrahydrofuran (THF, a good solvent) and injected into water (a poor solvent) with sonication. We measured the absorption and PL spectra of DCeTDCS in mixed solvents by varying the water to THF ratio. Unexpectedly, the three absorption spectra, respectively for 0%, 20%, 40% water fraction, show a blue-shifted band at 290 nm from the reported DCS band in solution ($\lambda_{\text{max}} = 390$ nm), and even more similar to that of almost parallel H-aggregated DCS (Figure 1c).^[13a] This shift may originate from the low solubility of DCeTDCS in most of the solvents. It looks like DCeTDCS is fully soluble in THF after heating. However, it seems that the DCeTDCS forms tiny parallel H-aggregates composed of several dye molecules in THF solution or in the slight presence of water. On the contrary, the absorption band shifted to 325 nm in the 60% water fraction, which indicates DCeTDCS is forming an antiparallel H-aggregation,^[13a] and, simultaneously, the spectra had a tail at longer wavelengths indicative of Mie scattering and the formation of a NP suspension (Figure 1b,c). Accompanying the absorption changes at 60% and 80% water fraction, the PL spectra were drastically enhanced as well as red-shifted from 430 to 515 nm (Figure 1d; Figure S1,

Supporting Information). Accordingly, the PL quantum yield was improved to 58% at 80% water fraction, while it was 0.03% at a lower water fractions (0%–40%) (Figure 1e). These results demonstrate that DCETDCS is an appropriate AIE LP and is highly suitable for FON experiments.

Laser-ablative synthesis of DCETDCS FONs was performed using femtosecond laser ablation, which has been successfully used primarily for the fabrication of metal and semiconductor nanoparticles. To prepare targets for laser-ablative experiments, we first dissolved and heated DCETDCS samples in THF. Then, the obtained solution was drop cast on a glass substrate and dried in vacuum for 12 h to form a thin film. Finally, the glass substrate with the film was placed on the bottom of a 20 mL glass vessel filled with 7 mL of deionized water (18 M Ω cm at 25 °C). The ablation of the films was carried out using radiation from a Yb:KGW femtosecond laser (1025 nm, 450 fs FWHM pulse duration, 100 μ J, 10 kHz repetition rate), which was focused with a 75 mm lens on the surface of a target (Figure 2a). The target was constantly moved in the focusing plane with the speed of 2.5 mm s⁻¹ to avoid ablation from the same area on the target while keeping the same thickness of the liquid of about one centimeter above the target. The laser ablative synthesis was done during 15 min to obtain relatively high concentrations of colloidal material. As shown in Figure 2b, the resulting colloidal solutions exhibited similar emission spectra as DCETDCS FONs obtained by the reprecipitation method in 80% water, which indicate the LPs are forming an antiparallel *H*-aggregation in the laser-fabricated FONs as well.

The size of the resulting nanoparticles obtained through laser ablation and reprecipitation was determined by atomic force microscope (AFM) measurements using an NT-MDT Ntegra Prima scanning probe microscope. Samples were prepared by drop-casting 10 μ L of a colloidal solution of DCETDCS on a borosilicate glass substrate and subsequent drying at ambient conditions for several hours. The microscope was configured to operate in a tapping (noncontact) mode with NSG10 probe (10 nm tip curvature radius, 10 N m⁻¹ force constant) to reduce force impact on the sample. A typical AFM image of the dried DCETDCS FONs was shown in Figure 2c and Figure S2 (Supporting Information). The ablated material is mostly composed of small nanoparticles that tend to aggregate during the drying process. The lateral size measurements of FONs obtained by AFM are not accurate because of the large size of the AFM probe, and thus cannot be used to assess their size. However, the size of the FONs can be assessed from the height measurement. The

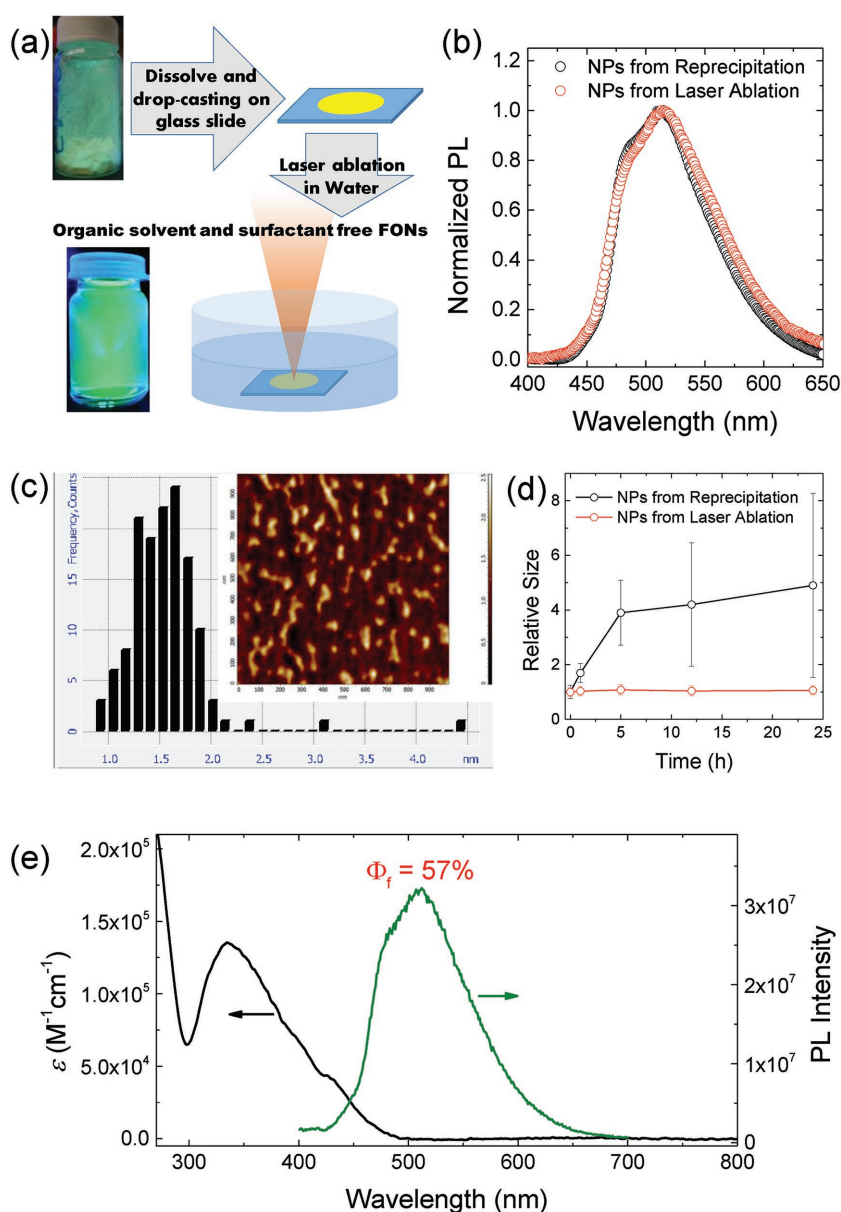


Figure 2. Fabrication of AIE FON by laser ablation. a) Schematic representation of laser ablation process of DCETDCS in water. b) Normalized PL spectra of DCETDCS FONs fabricated by reprecipitation and laser ablation. c) AFM image (inset) and height distribution of DCETDCS FONs by laser ablation. d) Change in size distribution of DCETDCS FONs fabricated by reprecipitation and laser ablation as a function of time after their fabrication. e) Molar extinction coefficient and photoluminescence quantum yield of the laser ablated DCETDCS FONs.

average size of the DCETDCS FONs obtained from the AFM data was about 1.6 nm (Figure 2c).

We evaluated the suspension stability of the FONs fabricated by both reprecipitation and laser ablation by tracking the dependence of the particle size distribution on time after their fabrication. The size distribution of the laser ablated FONs remained unchanged for 24 h while the size of reprecipitated FONs grew to about double of the initial size in 1 h, and continued to grow to form macroscopic precipitates after 5 h (Figure 2d). In addition, the laser ablated FONs exhibited very high suspension stability without any visible precipitation for about 6 months.

The brightness of photoluminescence by a material is defined by the product of photoluminescence quantum yield (PLQY) and light absorbance of the material.^[15] To determine the brightness of laser ablated AIE FONs, we estimated the molar concentration of the FONs based on the particle size in AFM image,^[16] and then measured their molar extinction coefficient. The scattering baseline originated from the interaction between light and the nanoparticles was subtracted using the adjacent-averaging method. The molar extinction coefficient at 337 nm was $1.35 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$, which is comparable to that of quantum dots (Figure 2e). The nonconcentration quenching feature of AIE LPs allows to ultimately increase the concentration of LPs in nanospace to 100% and, accordingly, the extinction coefficient is improved even with the tiny size. The PLQY of the laser ablated FON, 57%, was almost the same as with the reprecipitated FON (Figure 2e; Figure S3, Supporting Information). Both of the parameters (the PLQY and the molar extinction coefficient of the laser ablated AIE FON) approach those of quantum dots,^[15] enabling us to claim that the brightness of the DCETDCS FON is similar to that of quantum dots. The superior features of laser ablated DCETDCS FONs (i.e., solvent and surfactant free fabrication process; brightness by AIE; narrow size distribution; and high suspension stability) encouraged us to explore their bioapplications.

To demonstrate one possible biological application of these AIE FONs, we incubated the FONs with microglia cells and observed photoluminescence signal from the cells after washing off unbound and nonincorporated FONs. Initially, we did not detect any intracellular emission upon 358 nm excitation (Figure 3d), indicating that the FONs failed to be incorporated into the cell. To determine why there was poor cellular uptake, we measured the surface charge (zeta potential) of the AIE FONs and recognized that the surface has a highly negative charge ($-54.87 \pm 1.98 \text{ mV}$), apparently resulting in a severe repulsion

from the negatively charged cell membrane (Figure S4, Supporting Information). We then modified the surface of the FONs with a cationic biopolymer, chitosan,^[17] by electrostatic interaction (Figure 3a). The pristine FONs were added slowly to an aqueous solution of chitosan, and the mixture was stirred for 5 h at room temperature. With this surface modification, the average size of the FONs increased to 4.45 nm (Figure S5, Supporting Information) and the zeta potential changed to $43.2 \pm 2.2 \text{ mV}$ (Figure S4, Supporting Information). With the chitosan coated FONs, we observed bright emission in the cytosol of microglia (Figure 3d). Next, to investigate the impact of the AIE FONs on cell viability, we evaluated the cytotoxicity of pristine FONs and the chitosan coated FONs using the methyl thiazolyl diphenyltetrazolium bromide (MTT) assay (Figure 3b). The pristine FONs did not show any noticeable toxicity within 24 h, attributable to negligible interaction and cellular uptake of the FONs. In comparison, the chitosan coated FONs demonstrated a slight toxicity above $6 \times 10^{-6} \text{ M}$ following a 24 h incubation. At the same time, the nontoxic concentration, $2 \mu\text{M}$, is far higher than that from another AIE FONs produced by reprecipitation method in the previous studies.^[12,16] It is worth noting that the acute cytotoxicity of our FONs is similar to quantum dots.^[4] Finally, we studied the potential effect of AIE FONs on the cell cycle progression. Here, the cells were incubated with chitosan coated FONs at concentrations of 2 and $6 \mu\text{M}$ for 24 h and mitotic cells were counted. Consistent with the MTT assay, we found a slight reduction in the number of mitotic cells at $6 \times 10^{-6} \text{ M}$, whereas there is no change at $2 \times 10^{-6} \text{ M}$ (Figure 3c). Cells treated with either concentration of FONs continued to divide, and, in the following 48 h, covered the entire surface of the culture dish, showing no difference from the control. These findings indicate minimal cytotoxicity of AIE FONs in microglia.

In summary, a new FON was fabricated by laser ablation of AIE LP to obtain bright emission without the need of any

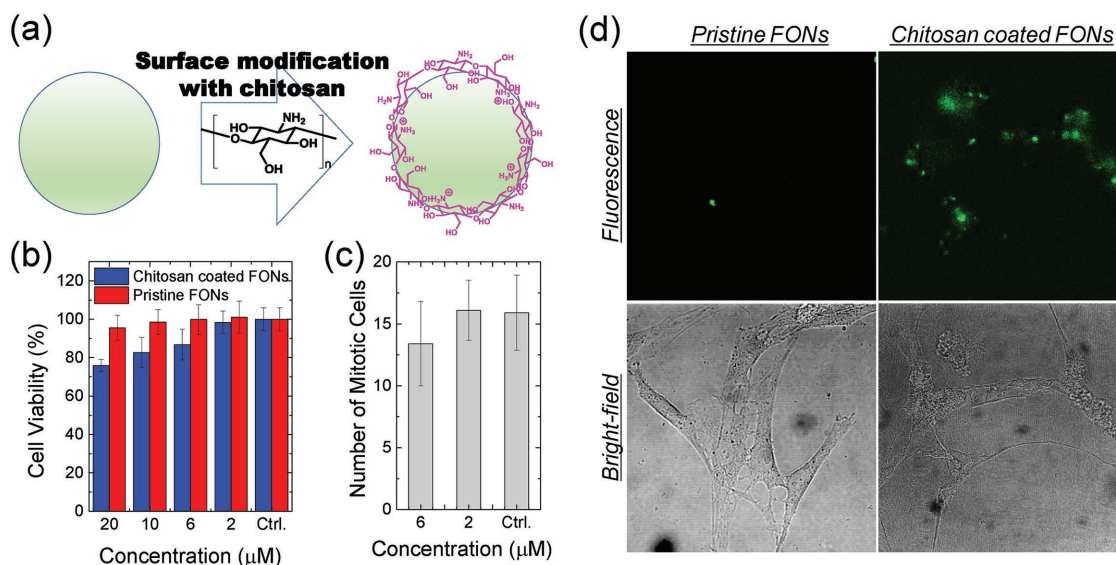


Figure 3. DCETDCS FONs in bioapplication. a) Schematic representation of the surface modification of DCETDCS FON with chitosan. b) Cell viability of microglia at various concentration of pristine and chitosan coated FONs for 24 h. Chitosan coated FONs were prepared by mixing of 0.01 mg mL^{-1} chitosan and various concentration of FONs. The same concentration of chitosan (0.01 mg mL^{-1}) was added to the control as well. c) Number of mitotic cells in the field of view at various concentration of chitosan coated FONs for 24 h. d) Bright-field (bottom) and fluorescence (top) images of microglia incubated with pristine (left) and chitosan coated (right) FONs for 16 h.

additional organic solvent and surfactant. For this, a new AIE LP was designed, synthesized, and shown to demonstrate strong AIE activity. It exhibited green-yellow emission with a high PLQY (up to 58%) in the aggregated state. Laser ablation of this material in water produced very stable FONs suspension without any organic solvent and surfactant. Their suspension stability was far superior to that of reprecipitated FONs. By measuring the molar extinction coefficient and PLQY, we determined that the brightness of this FON is comparable to that of quantum dots. With this bright and stable AIE FON, we successfully demonstrated a cellular imaging application using a simple surface modification to increase the zeta potential of the FONs with chitosan, which dramatically improved cellular uptake. This strategy of fabricating FONs by laser ablation of pure AIE LPs can be a promising direction to produce brighter and less toxic nanoparticles for biomedical applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

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